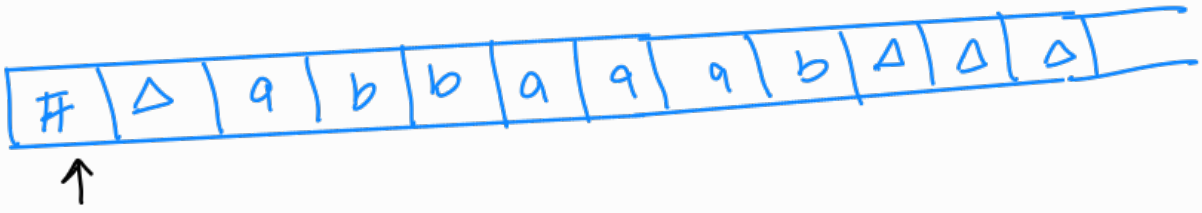
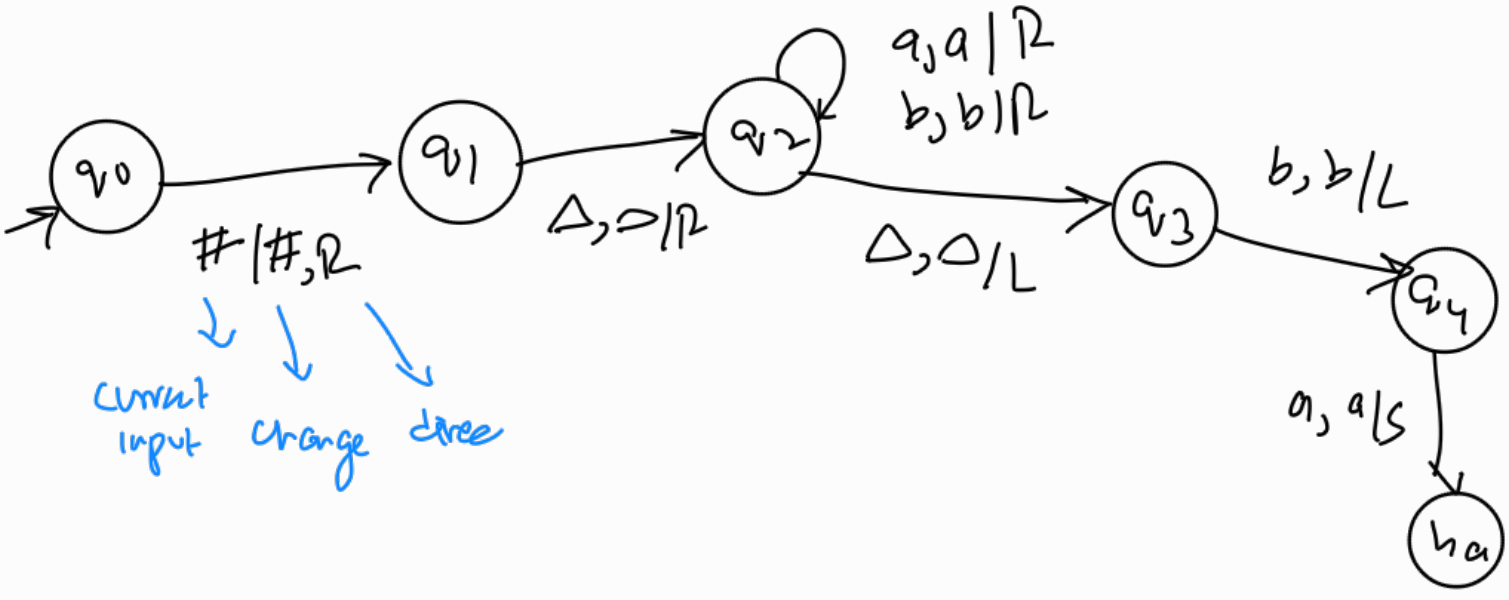
Main goal is to read input successfully →

Head ↑ Right/write → can Read and modify

$L = \{ w \mid w \in \Sigma^* \}$ w ends with ab} e.g abbbab

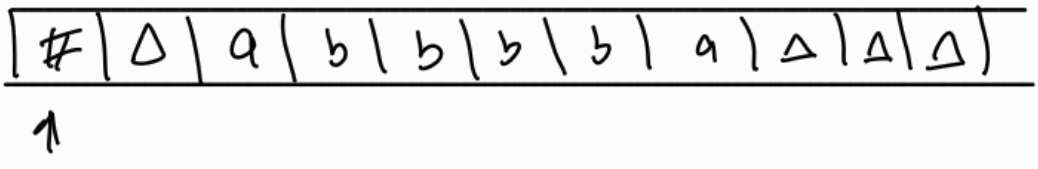


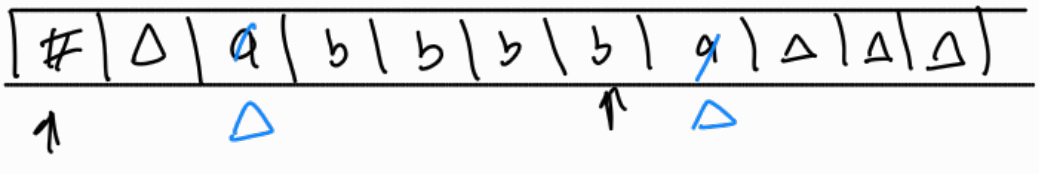


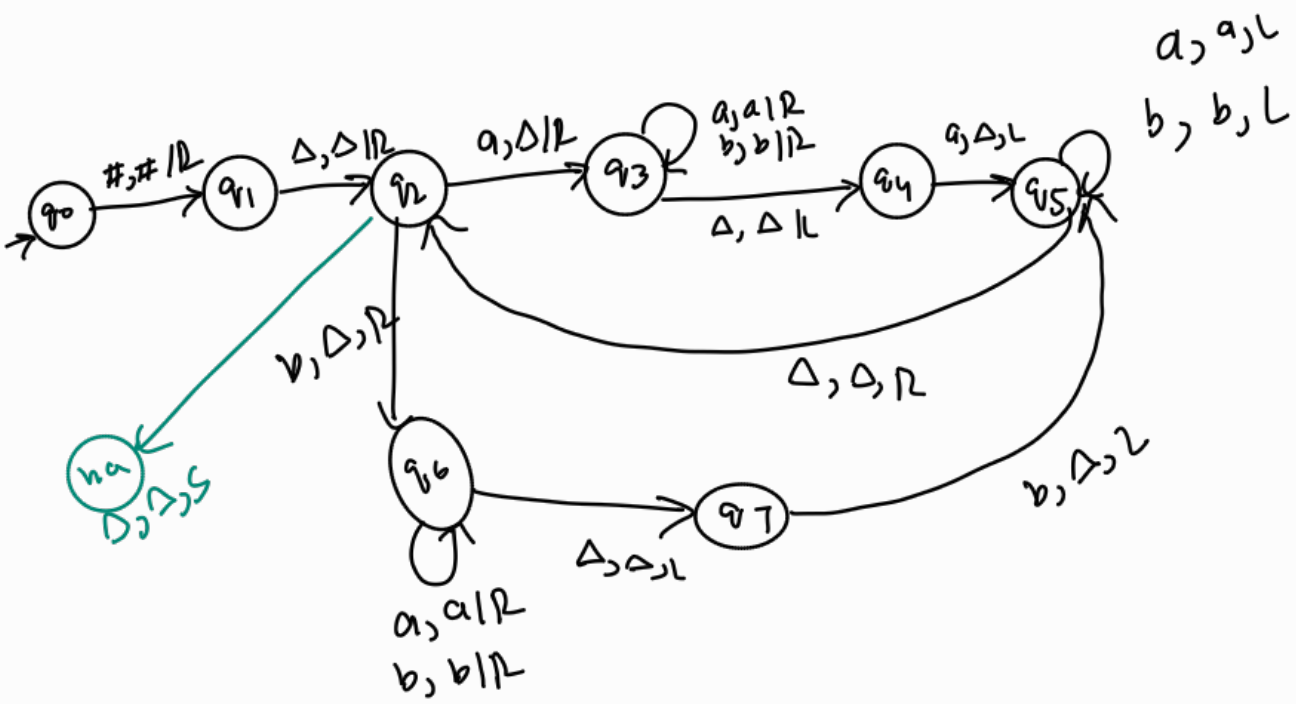
haltaccept → Valid input | Stationary

haltreject → Invalid input | Stationary

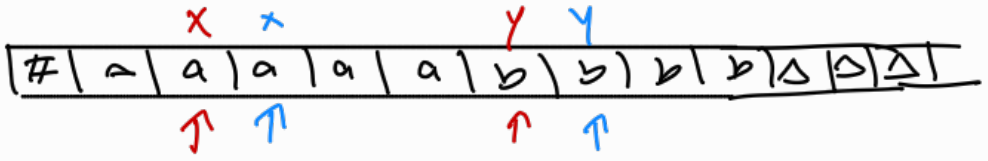
$L = \{ w \mid w \in \Sigma^* \}$ w is palindrome}



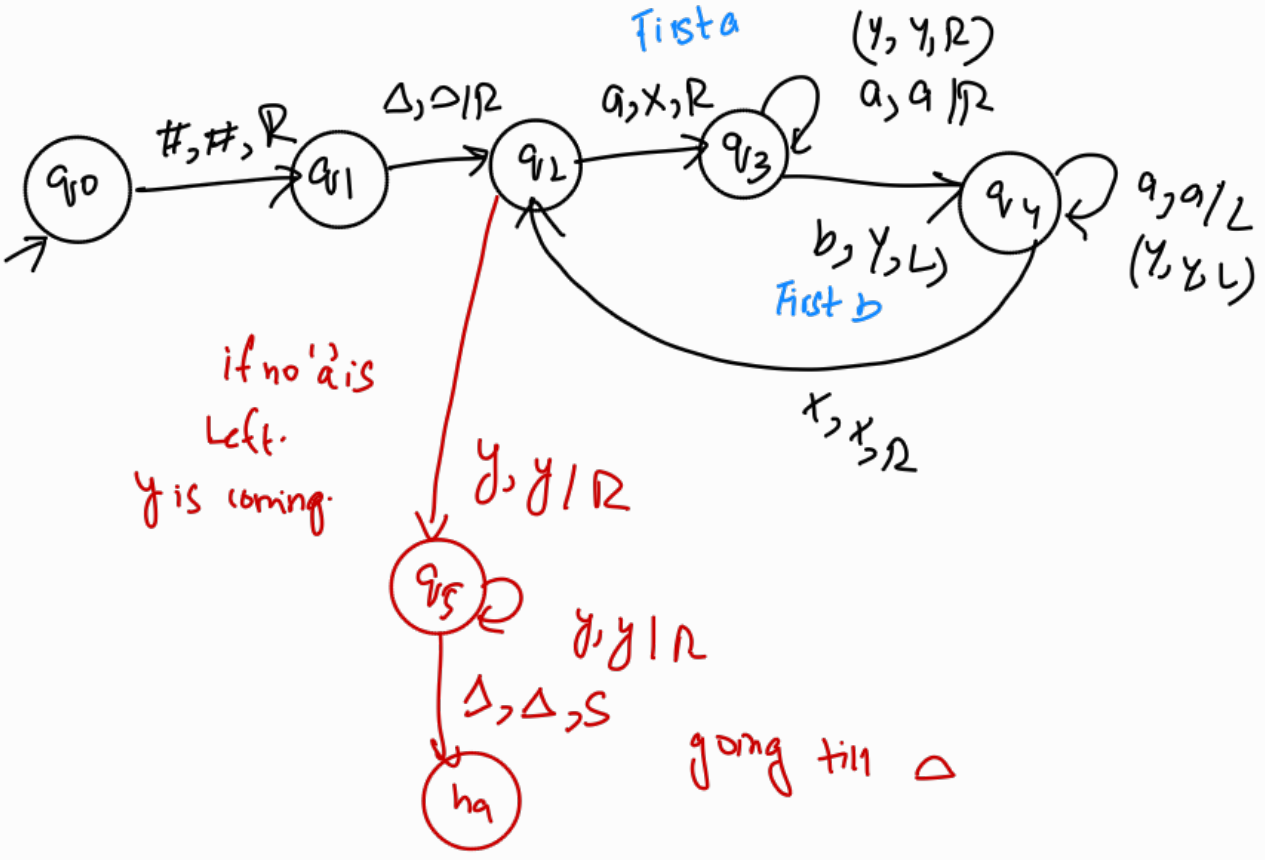




L = { equal no of 'a' and equal no of 'b' }

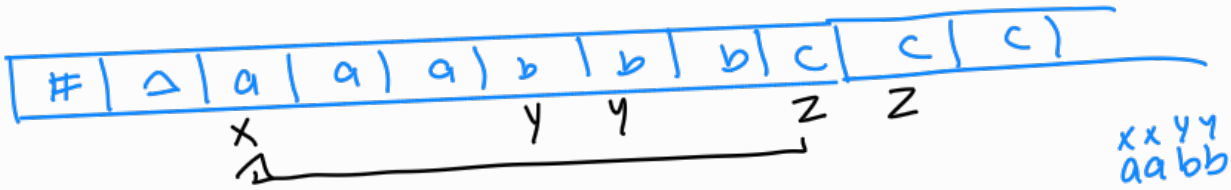


For first a mark X and then for first b mark Y do this for rest.

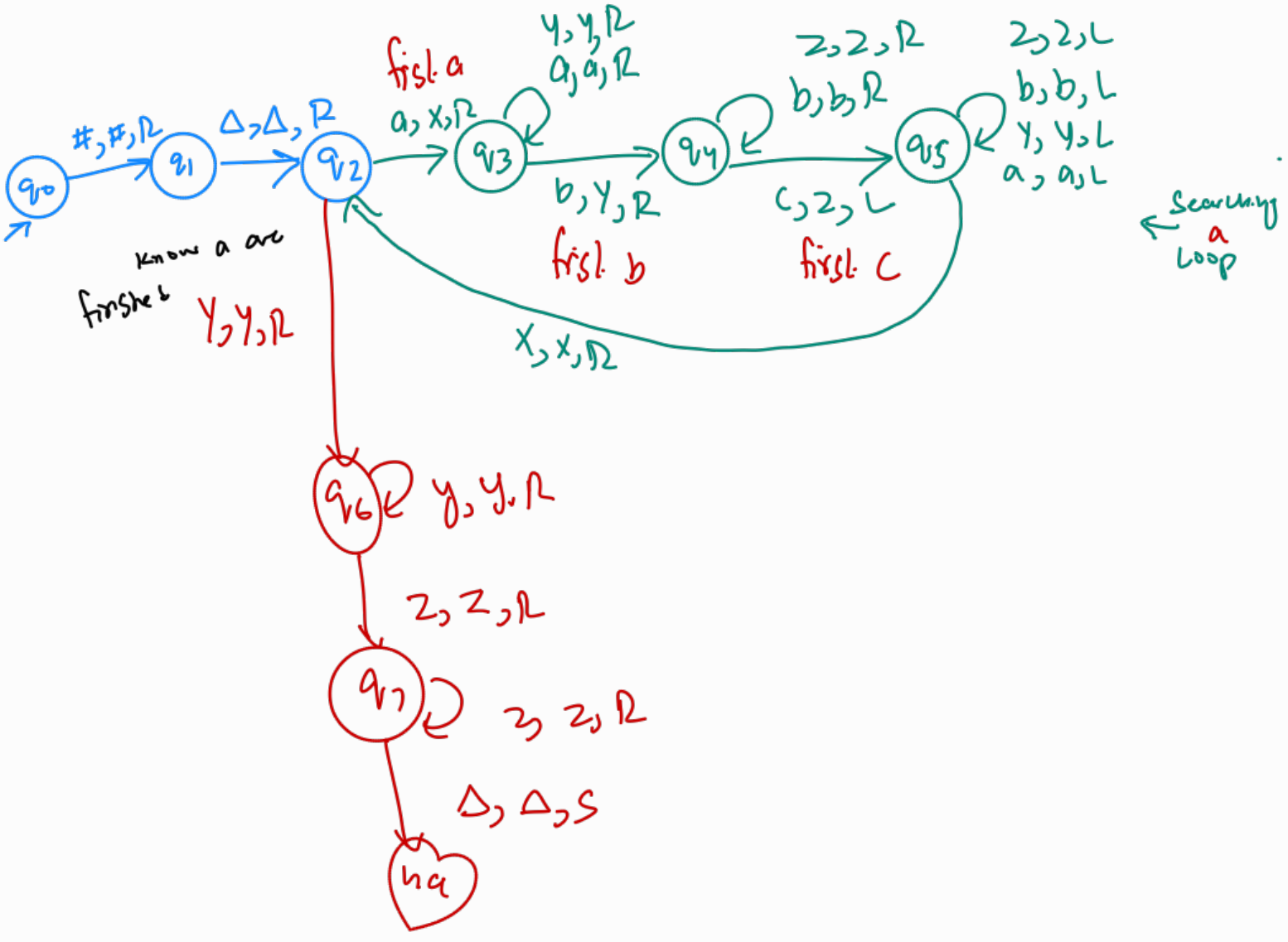


Design T.A for $(L = a^n b^n c^n \mid n \geq 1)$

(aaa bbb ccc)



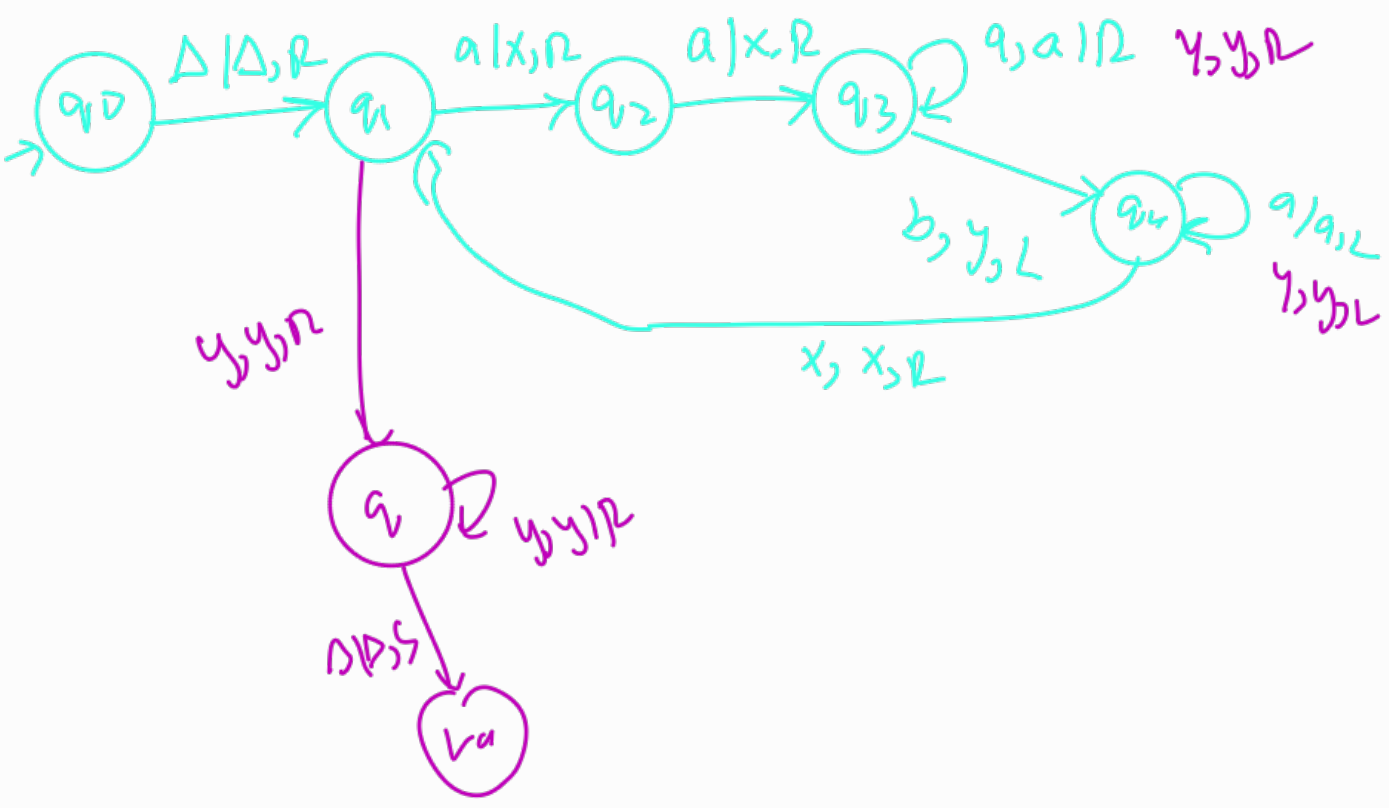
xx yy zzL
aa bb ccc (Invalid)



Design a turning machine for $L = \Sigma a^{2n} b^n \mid n \geq 1$

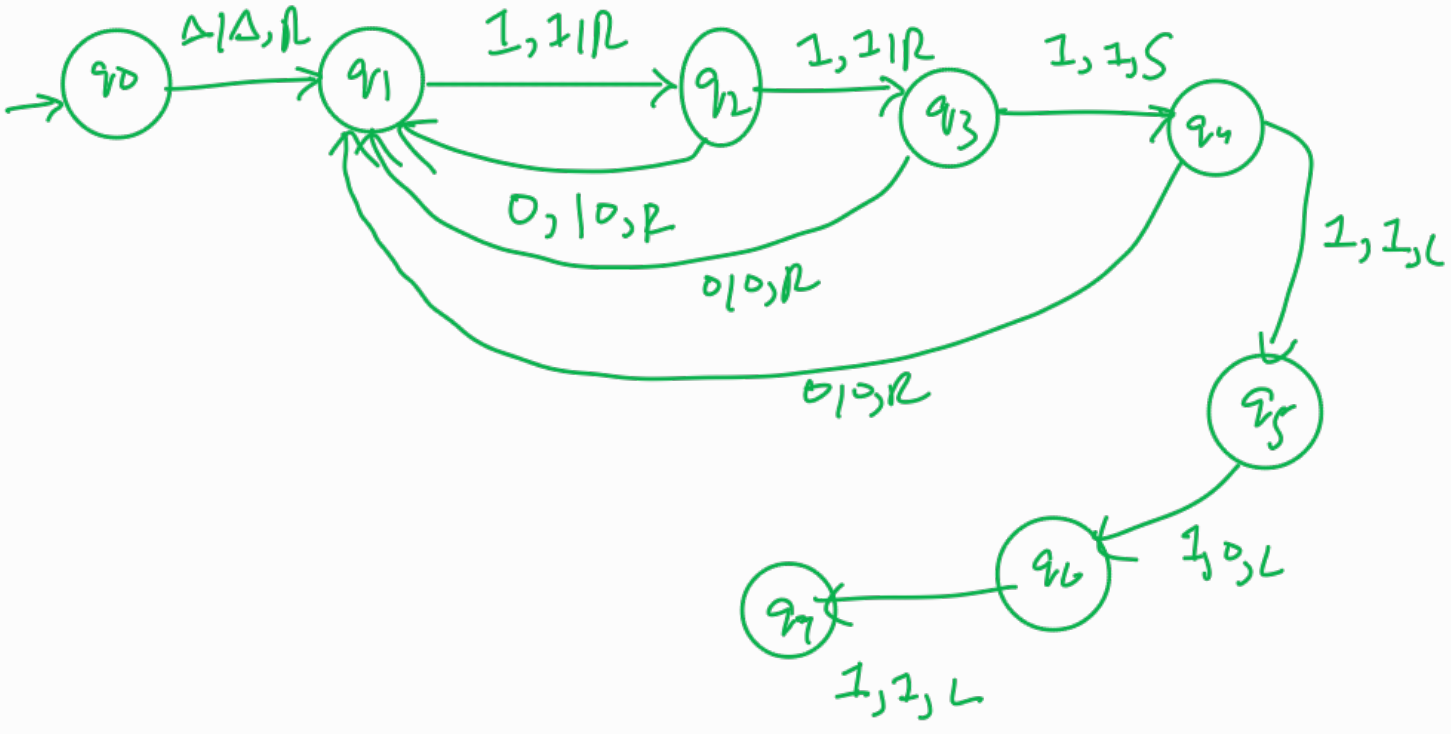
aa aa bb
x x Y Y

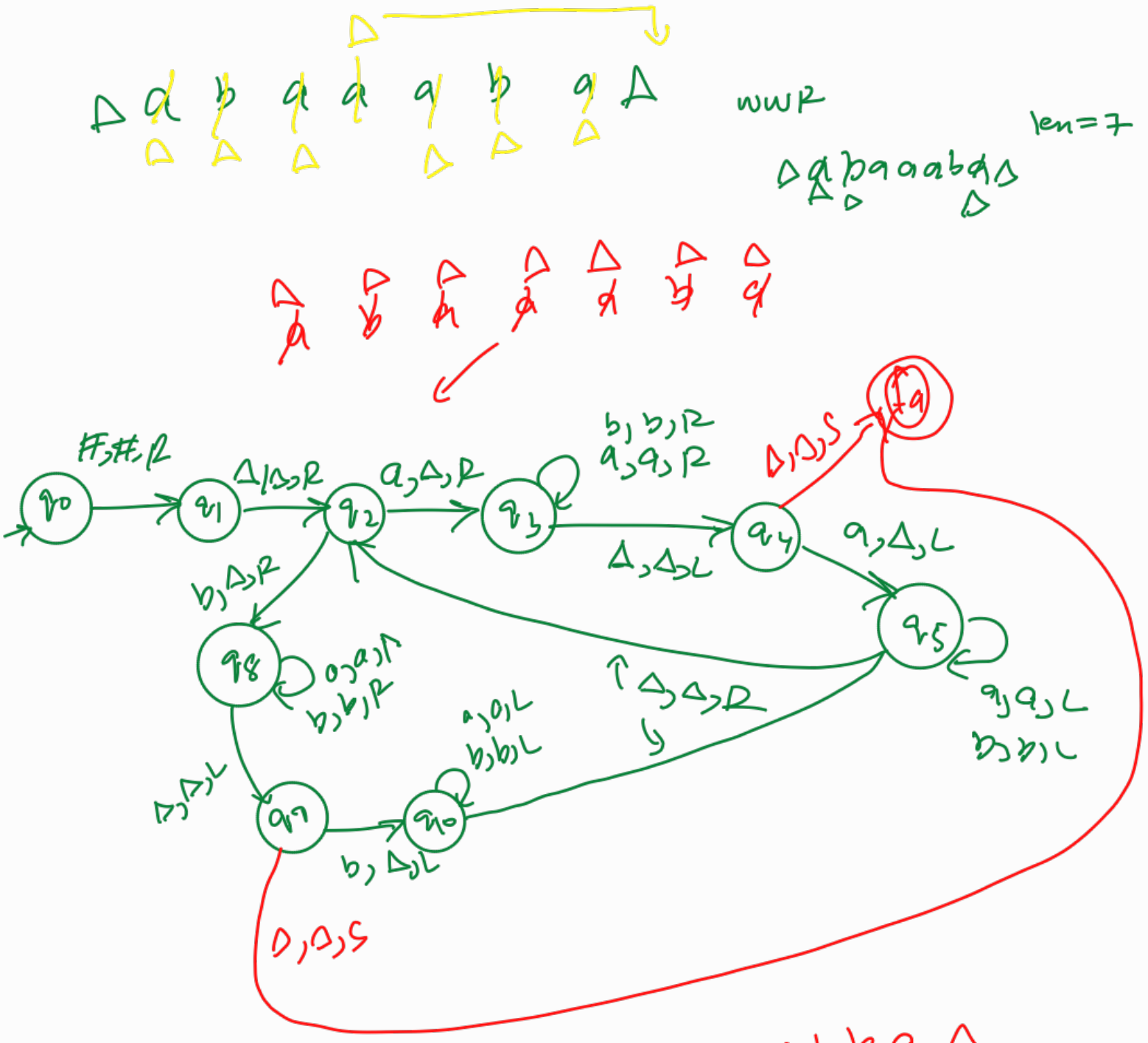
Read two a's and replace them with X a



Design a turning machine that replaces all occurances of 111 by 101 from sequence of 0's and 1's

1 0 1 1 0 1 0 0 1 1 1 0 1 1 1 1 1 0





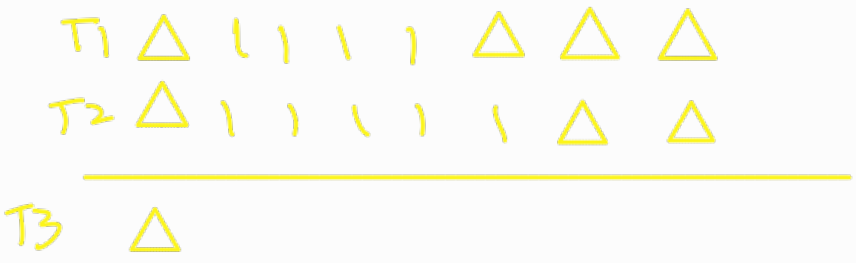
ΔabbaΔ

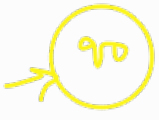
even palindrom will not be accepted.

TM of a adder

a = 4 b = 5 a+b = 9

a = 1111 b = 11111 111111111

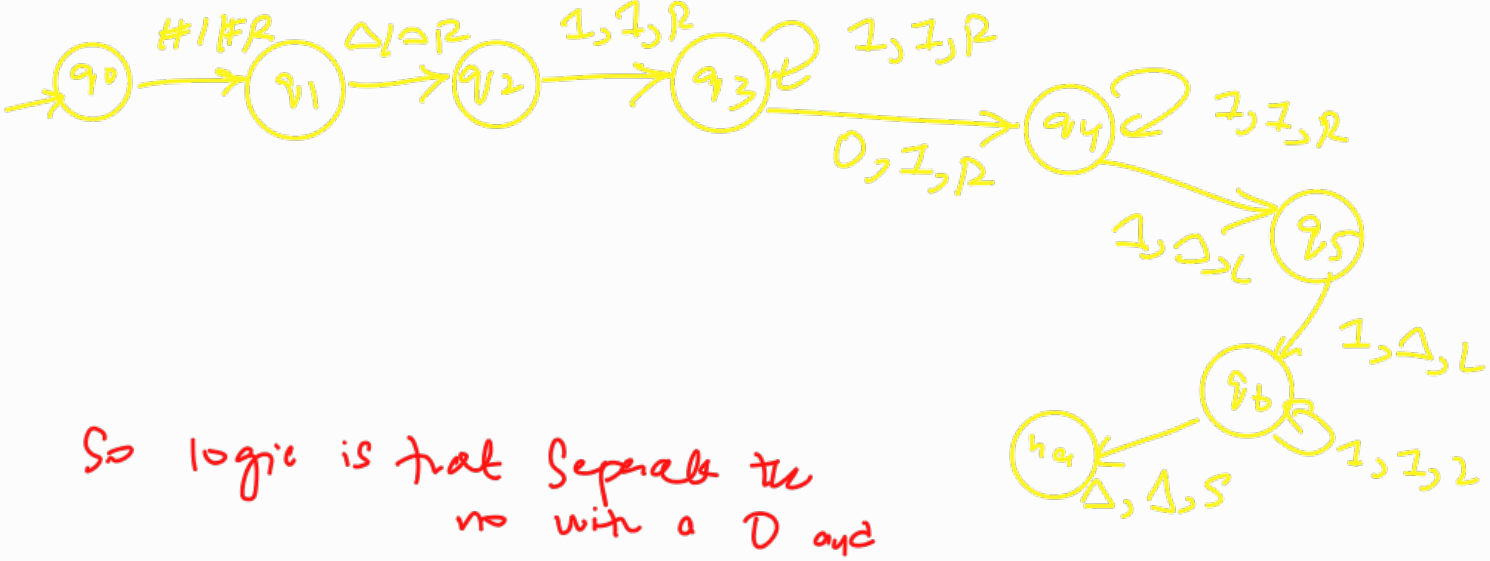




(this was done on a single tape)

zeros as a separator.





So logic is that Separate the no with a 0 and change that 0 into 1 while reading from L→R and for the last 1 convert it into Δ.

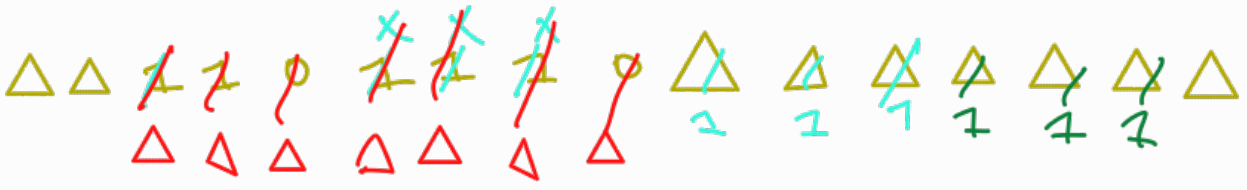
TM for multiplication x=2 y=3 x*y=6
Single tape 11 111



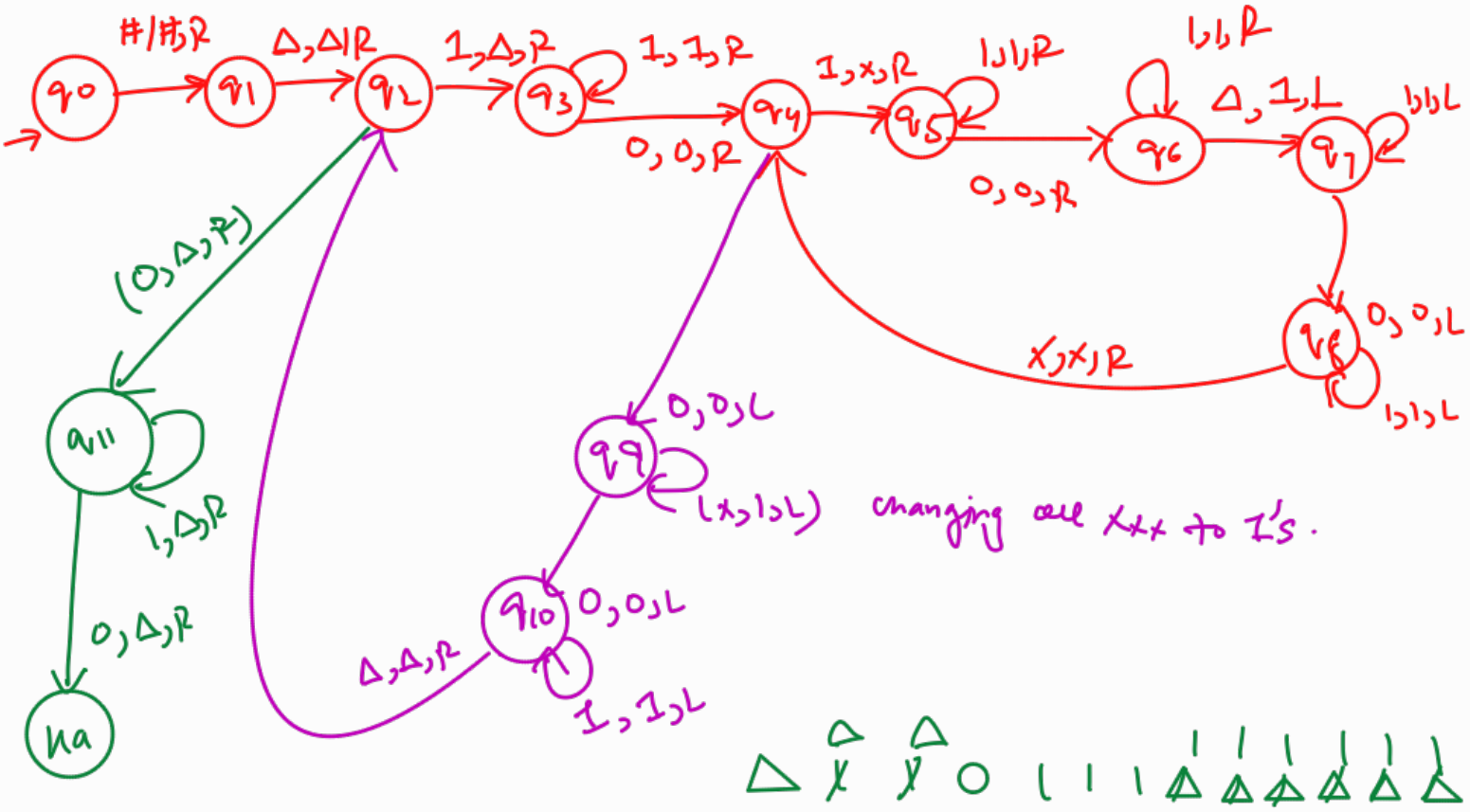
Read the first 1 and convert it to Blank. Place the three ones now into Blanks by converting it into X.



After coming Back Reconvert the X's into 1's for the new



After all the 1's Have place at the Right side make sure there are only 6 1's.



$\Sigma = \{1\}$ Input x → many input

Divide x by 2 and Store Quotient

$6 \div 2$

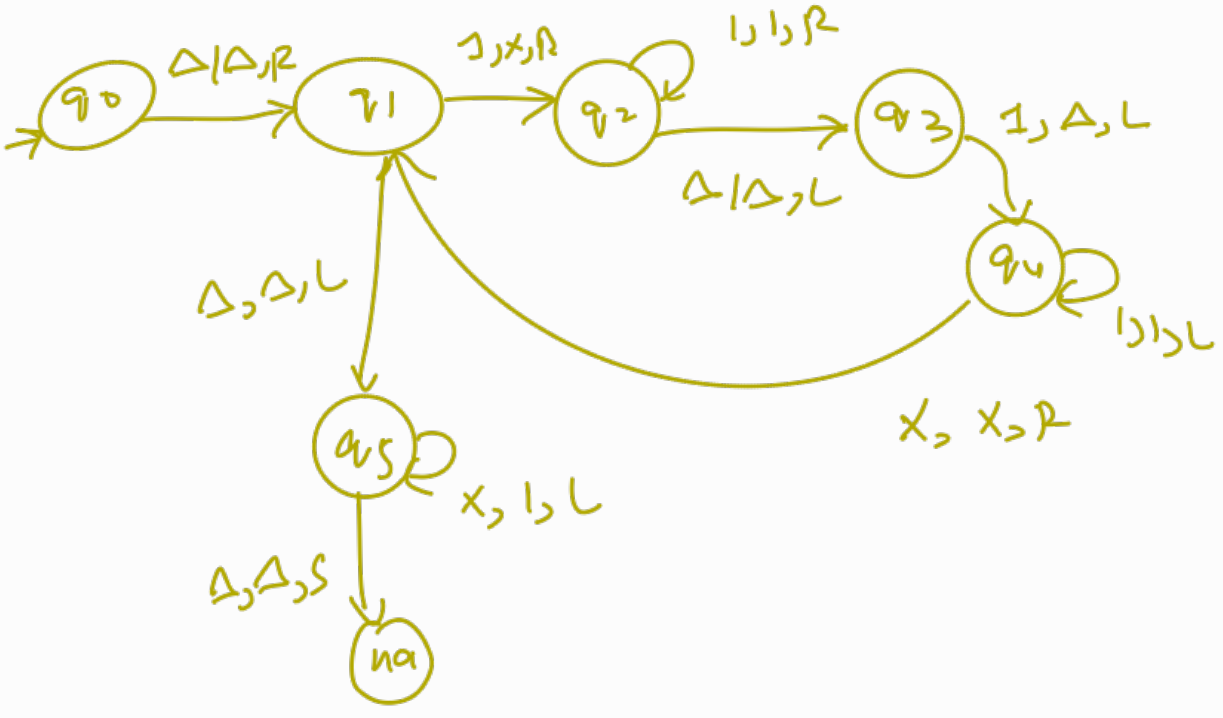
$Q \to 3$

Δ 1 1 1 1 1 1 Δ

x x x Δ Δ Δ

Storing only the quotient

For every 1 marked it write nue



$\frac{7}{2}$ $Q = 3$

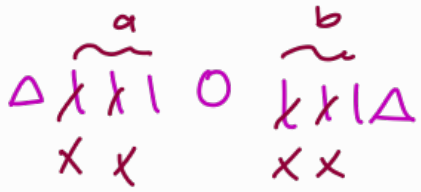
Δ 1 1 1 1 1 1 1 Δ

x x x x Δ Δ Δ

Δ

Answer

TM as a comparator:-



a = b
a > b
a < b

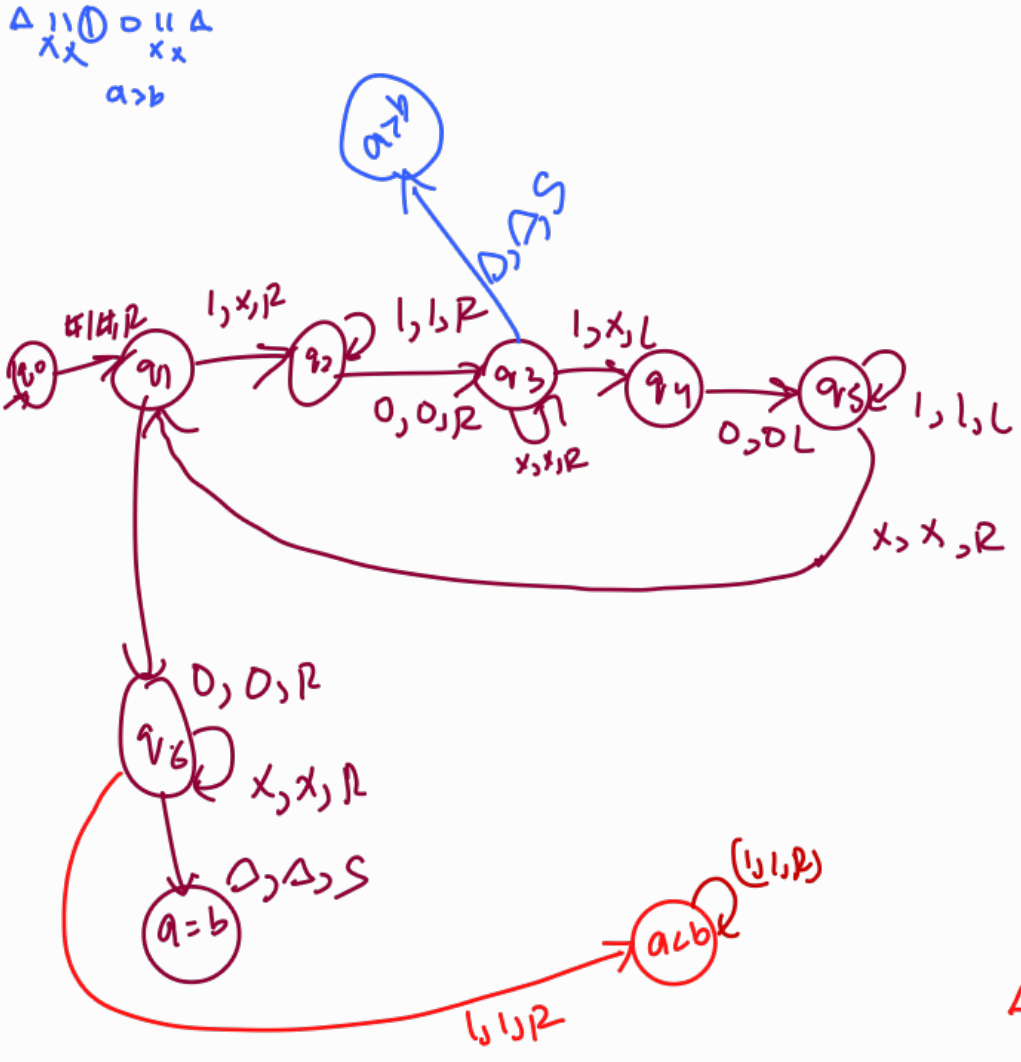


a < b



a > b

0 as a Separator

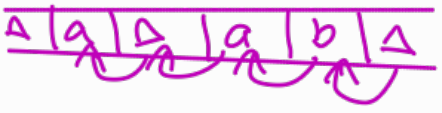


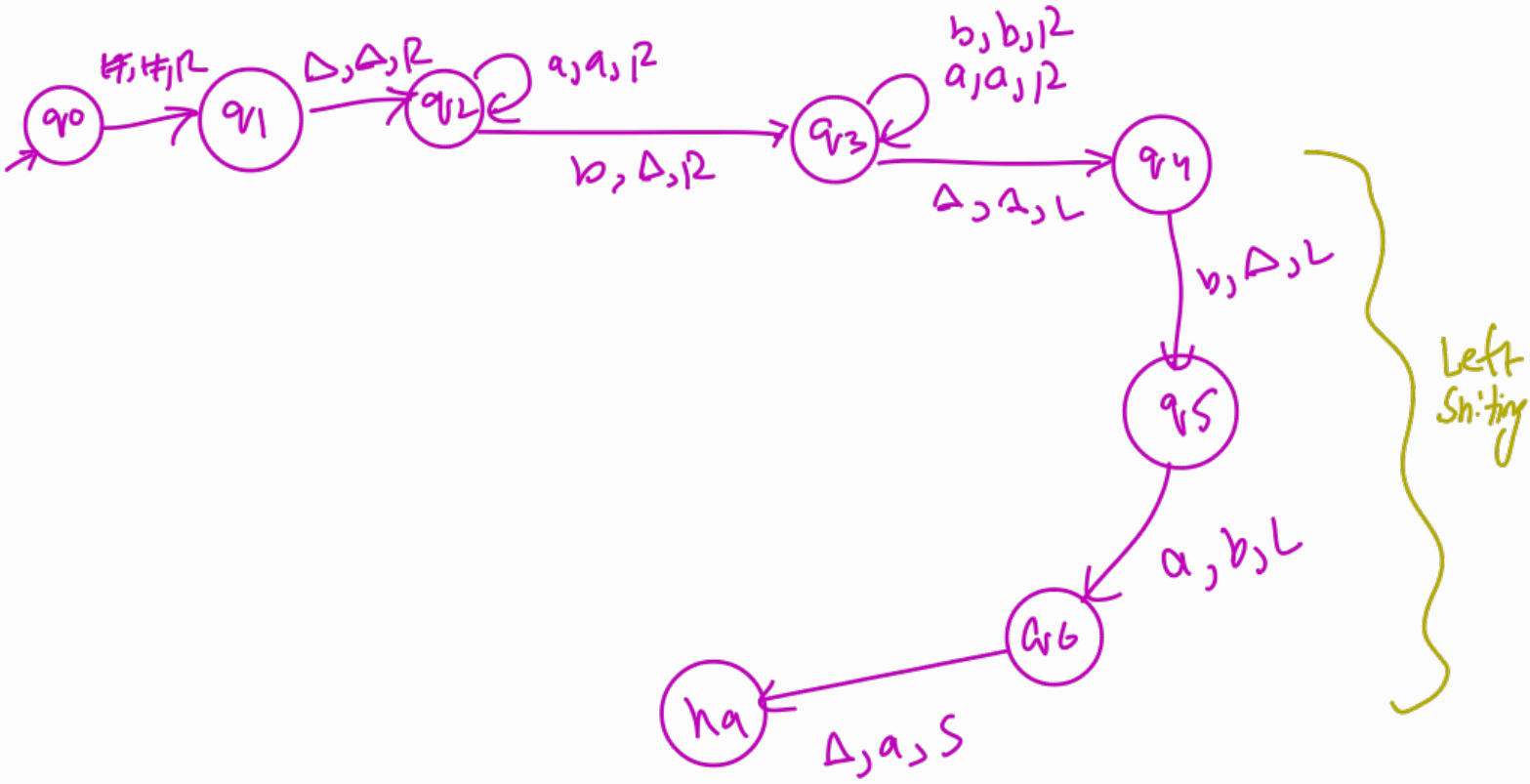


TM to delete a Symbol



- Replace the symbol you want to delete by Δ.
- keep moving to R until you get Δ
- keep shifting L until you get Δ.





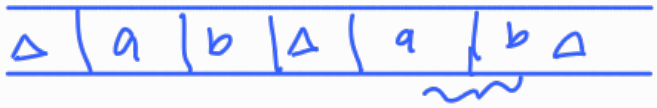
Δ a b a b Δ

Δ a Δ a b Δ

Δ a a b Δ Δ

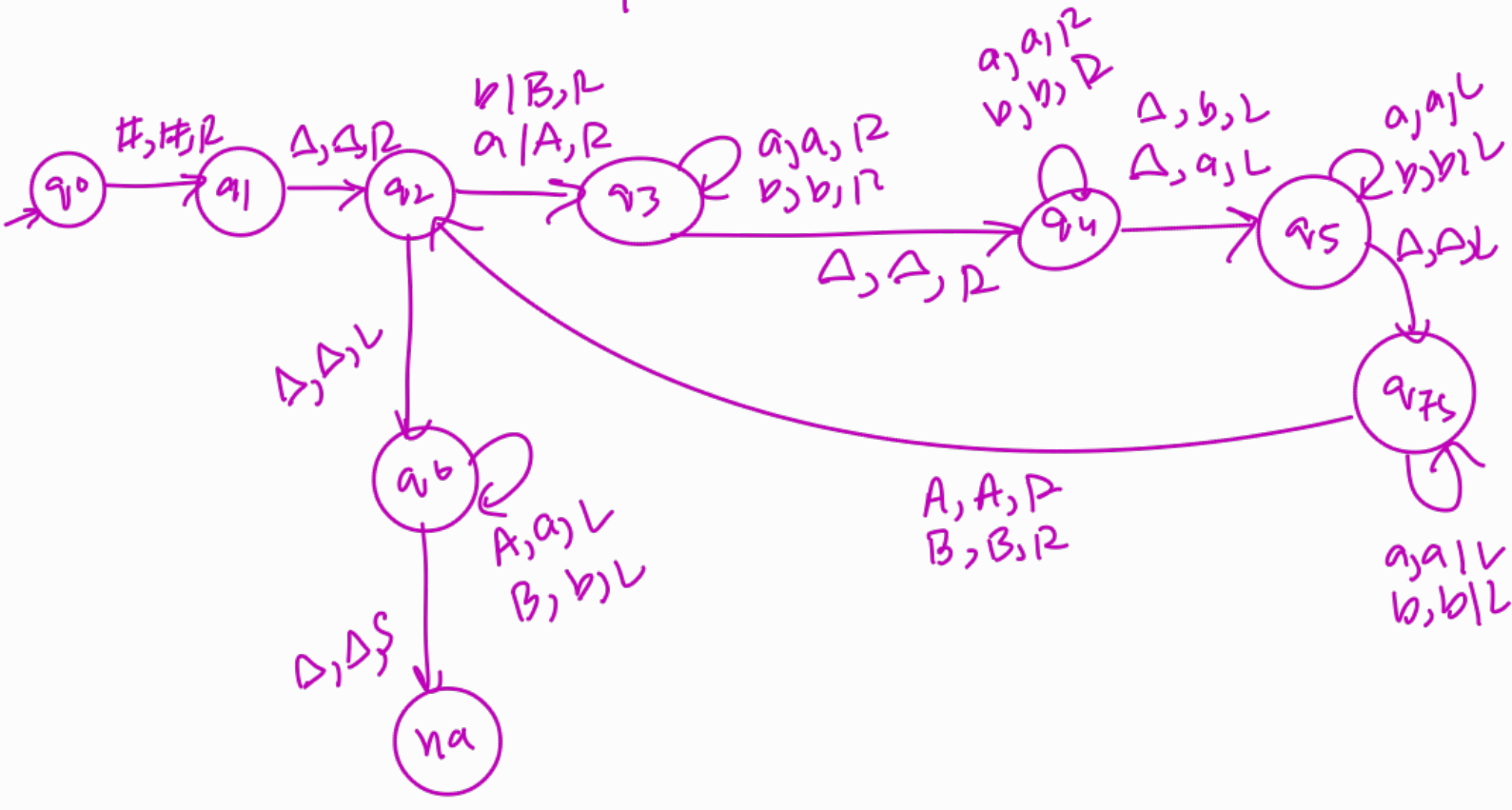
TM to copy a String:-



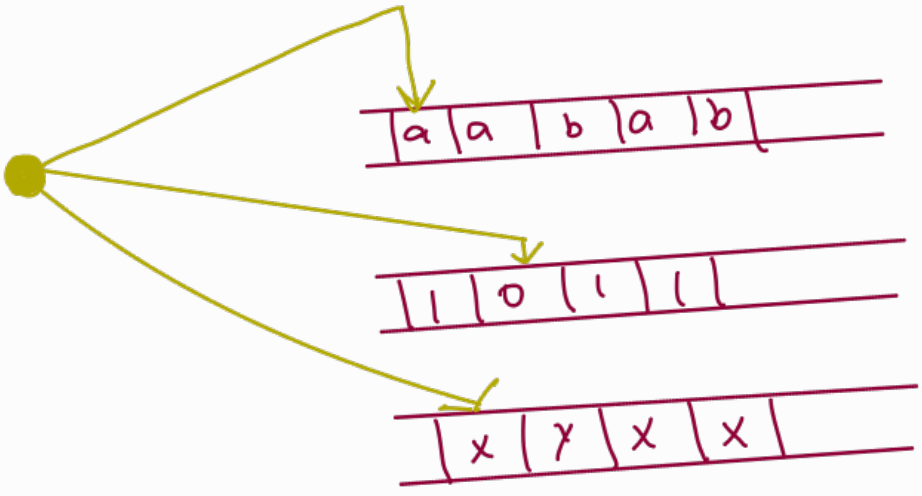


copied





Multitape turning machine → every multipe tape turning machine has an equivalent Single tape turning machine





a0x | b1y | RPR

↓

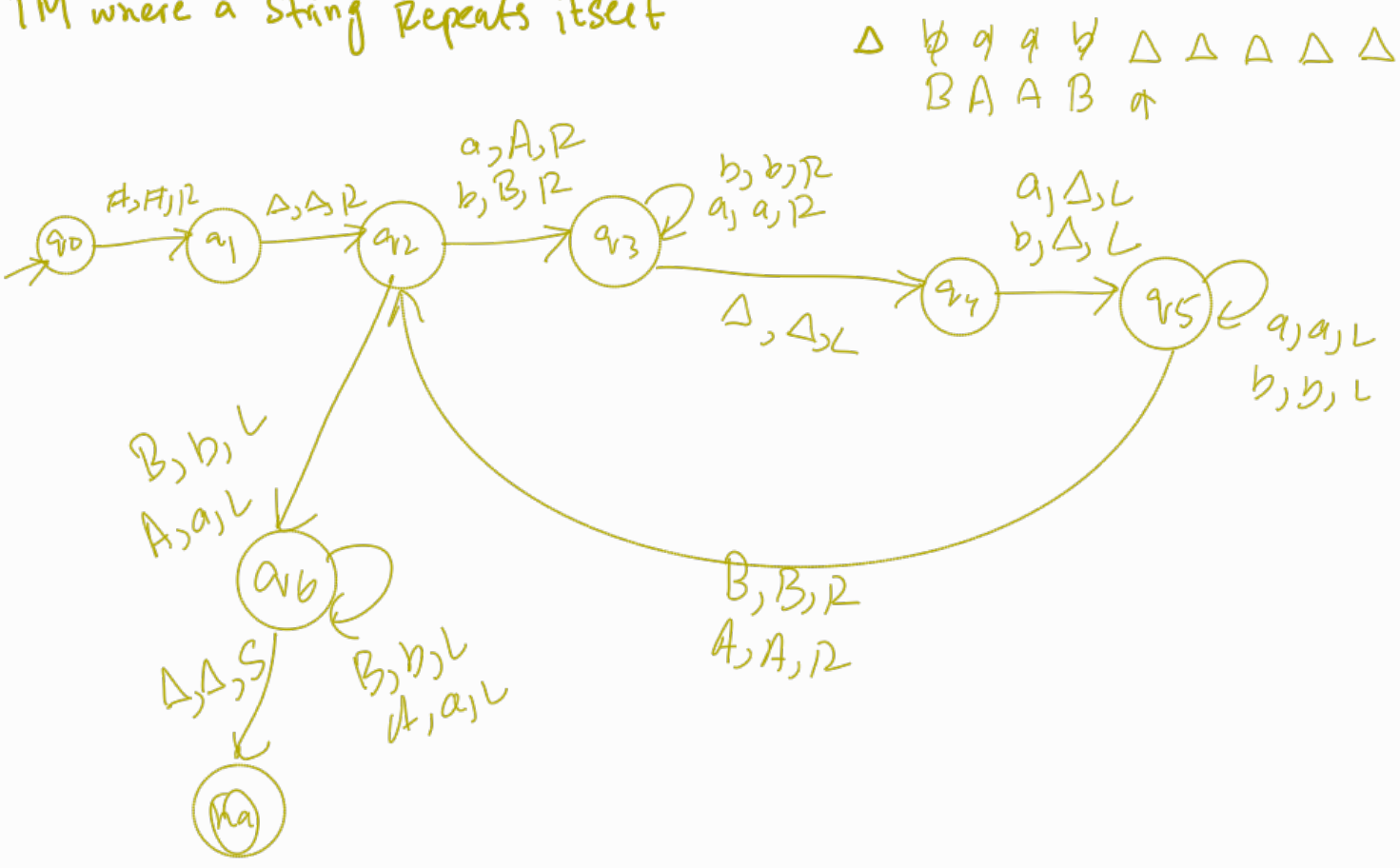
Replace a by b
0 by 1
x by y

and move all three heads to R.

Representing in Single tape::



TM where a String Repeats itself



T.M for Reversing a String:.

Δ a b ab Δ Δ Δ Δ

Δ Δ Δ Δ Δ b a b a Δ

logic → traverse the end of the string and Replace b by B. and inplace of Δ write b.

Δ a b a b Δ Δ Δ Δ

X X X b a b

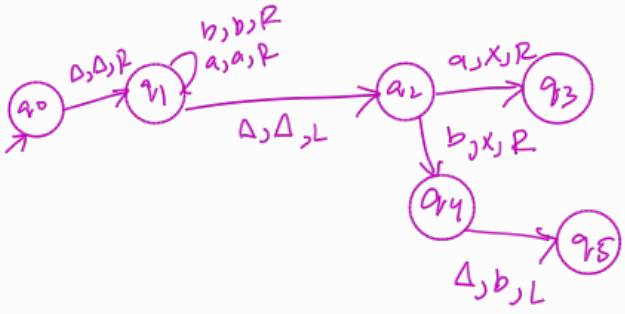
a b a X b Δ Δ

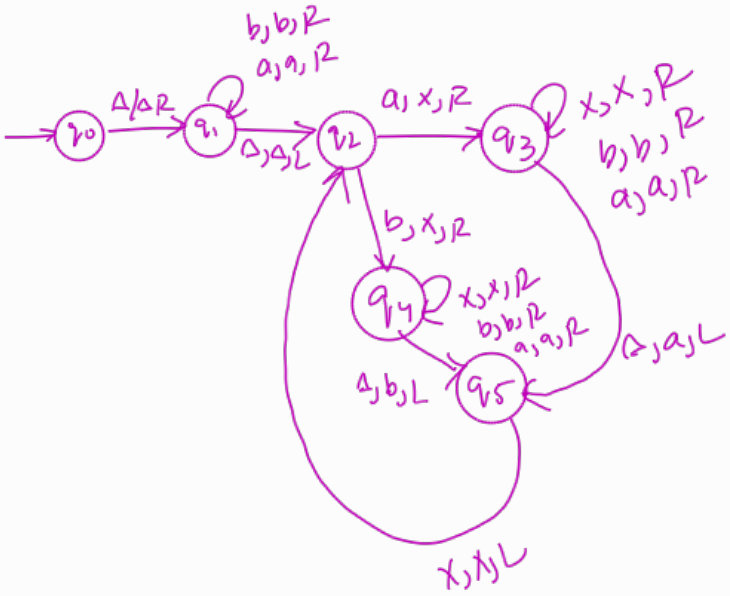
X ↑

then go back again to the string:.

a b b b

x





Δ a b a X Δ Δ Δ Δ

↑ b